

Dr. Babasaheb Ambedkar Technological University

Supplementary Examination – Summer 2024

Course: B. Tech. Branch: Civil Engineering Semester: V

Subject Code & Name: BTCVC502 Geotechnical Engineering

Max Marks: 60

Date: 03/07/2024

Duration: 3 Hrs

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve <u>any two</u> of the following.		12
A) Write the difference between 'residual' and 'transported' soils.	Understand/ CO1	06
B) One cubic metres of wet soil weighs 20 kN. If the specific gravity of soil particles is 2.70 and water content is 20 %, find the void ratio, dry density and degree of saturation.	Apply/ CO1	06
C) A saturated clay has a water content of 40% and a bulk specific gravity of 1.84. Determine the void ratio, porosity and specific gravity of particles.	Analyze/ CO1	06
Q.2 Solve <u>any two</u> following questions.		12
A) Define the following terms: A.1) Plasticity Index, A.2) Shrinkage Index, A.3) Consistency Index	Understand/ CO1	06
B) Enlist the different methods for the determination of water content recommended by IS code and explain one of the methods in detail.	Understand/ CO1	06
C) The dry unit weight of a sand sample in the loosest state is 13.34 kN/m ³ and in the densest state, it is 21.19 kN/m ³ . Determine the density index of this sand when it has a porosity of 33%. Assume the grain specific gravity as 2.68	Apply / CO1	06
Q. 3 Solve <u>any two</u> of the following.		12
A) Enlist the different methods to determine the permeability of soil and explain the one of them.	Understand/ CO2	06
B) A sample in a variable head permeameter is 8 cm in diameter and 20 cm high. The permeability of the sample is estimated to be 10×10^{-4} cm/s. If it is desired that the heading the stand pipe should fall from 24 cm to 12 cm in 3 min., determine the size of the standpipe which should be used.	Apply / CO2	06
C) A 1.25 m layer of the soil ($G = 2.65$ and porosity = 40 %) is subject to an upward seepage head of 1.85 m. What depth of coarse sand would be required above the soil to provide a factor of safety of 2.0 against piping assuming that the coarse sand has the same porosity and specific gravity as the soil and that there is negligible head loss in the sand.	Analyze / CO2	06

Q.4 Solve <u>any two</u> of the following.		12
A) Explain the direct shear test.	Understand/ CO2	06
B) A vane, 10.8 cm long, 7.2 cm in diameter, was pressed into a soft clay at the bottom of a bore hole. Torque was applied and the value at failure was 45 Nm. Find the shear strength of the clay on a horizontal plane.	Apply / CO2	06
C) Calculate the potential shear strength on a horizontal plane at a depth of 3 m below the surface in a formation of cohesionless soil when the water table is at a depth of 3.5 m. The degree of saturation may be taken as 0.5 on the average. Void ratio = 0.50; grain specific gravity = 2.70; angle of internal friction = 30° . What will be the modified value of shear strength if the water table reaches the ground surface ?	Analyze / CO2	06
Q. 5 Solve <u>any two</u> of the following.		12
A) Explain in detail the concept of compressibility of soils.	Understand/ CO2	06
B) State and explain the assumptions in 'Terzaghi's theory of 1D consolidation'.	Understand/ CO2	06
C) A retaining wall, 6 m high, retains dry sand with an angle of friction of 30° and unit weight of 16.2 kN/m^3 . Determine the earth pressure at rest. If the water table rises to the top of the wall, determine the increase in the thrust on the wall. Assume the submerged unit weight of sand as 10 kN/m^3 .	Apply / CO2	06

*** End ***